MATH329 · OPERATIONS RESEARCH

Midterm 462 Solutions

09/03/2025 · 30 marks

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01 Solution Q1: theory

a The special cases of the graphical method

1. Alternative optimal solutions
2. Unbounded solution
3. No feasible solution

b The two requirements to start simplex

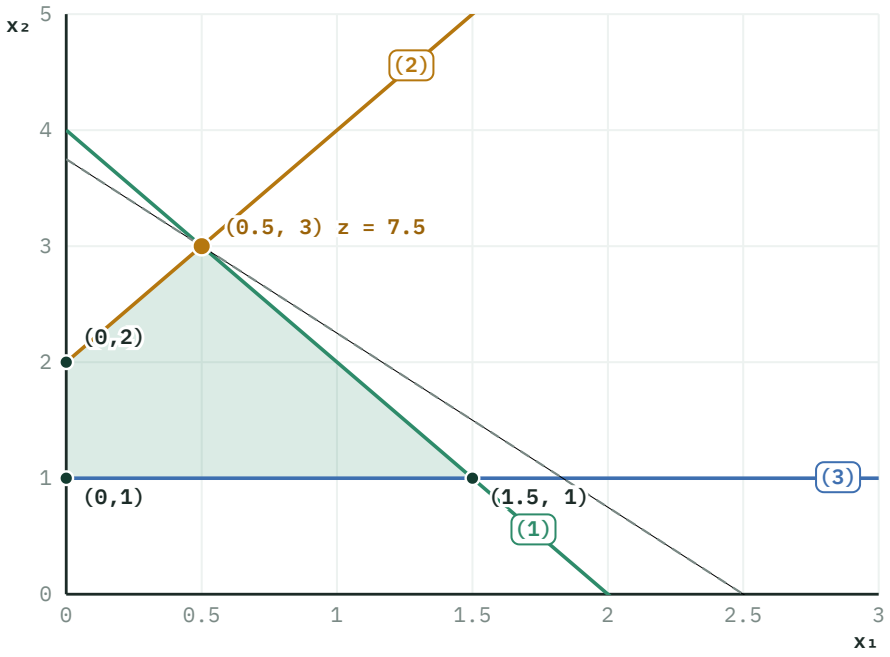
1. The coefficients of the basic variables (slack variables) form an **Identity matrix**.
2. In the objective function row, the coefficients of the basic variables are **zero**.

02 Solution Q2: graphical

1 Prepare the constraints

(1)	$2x_1 + x_2 = 4$	points (0, 4) and (2, 0)
(2)	$-2x_1 + x_2 = 2$	points (0, 2) and (−1, 0)
(3)	$x_2 = 1$	horizontal line

Constraints (1) and (2) hold at $(0, 0)$, so their region is on the origin's side.
Constraint (3) is $\geq$, so its region is above the line $x_2 = 1$.



2 The corner points

Corner	Where it comes from	$z = 3x_1 + 2x_2$
(0, 1)	(3) with the x_2 axis	2
(1.5, 1)	(1) with (3)	6.5
(0.5, 3)	(1) with (2)	**7.5**
(0, 2)	(2) with the x_2 axis	4

(1) + (2): $2x_2 = 6 \rightarrow x_2 = 3$ $\quad$ $2x_1 + 3 = 4 \rightarrow x_1 = 0.5$

Z = 7.5 at $x_1 = 0.5$ and $x_2 = 3$

03 Solution Q3: standard form

1 The methods 2 M

Graphical method and Algebraic method.

2 Standard form (6 M)

$4x_2 = -7 \quad \times (-1) \quad \rightarrow \quad -4x_2 = 7$

$x_2 = x_2^+ - x_2^-$ in every place x_2 appears

Maximum $Z = 8x_1 - 2x_2^+ + 2x_2^- + 0s_1 + 0s_2$	
S.to	$2x_1 + x_2^+ - x_2^- + s_1 = 12$
	$x_1 - 5x_2^+ + 5x_2^- - s_2 = 6$
	$-4x_2^+ + 4x_2^- = 7$
	$x_1, x_2^+, x_2^-, s_1, s_2 \geq 0$

Marking: two marks for the objective function and one for each of the four lines.

04 Solution Q4: simplex for a min problem

a Standard form (2 M)

Minimize $Z = x_1 - 2x_2 + 0s_1 + 0s_2$	
S.to	$2x_1 + 3x_3 + s_1 = 1$
	$3x_1 + 2x_2 - x_3 + s_2 = 5$
	$x_1, x_2, x_3, s_1, s_2 \geq 0$

The key writes the last line as $x_1, x_2, s_1, s_2, s_3 \geq 0$, which is a typo: the problem has x_3 and no s_3.

b Simplex `8 M`

The `Z` row comes from `z − x₁ + 2x₂ = 0`. The problem is min, so the entering variable is the most positive entry.

Basic	x_1	x_2	x_3	s_1	s_2	Sol.	Ratio
Z	−1	2	0	0	0	0	
s_1	2	0	3	1	0	1	–
s_2	3	**2**	−1	0	1	5	5/2
Z	−4	0	1	0	−1	−5	
s_1	2	0	**3**	1	0	1	1/3
x_2	3/2	1	−1/2	0	1/2	5/2	–
Z	−14/3	0	0	−1/3	−1	−16/3	
x_3	2/3	0	1	1/3	0	1/3	
x_2	11/6	1	0	1/6	1/2	8/3	

Iteration 1: the most positive entry is `2` under `x₂`. The `s₁` row has a zero coefficient, so it has no ratio, and the `s₂` row has ratio `5/2`. So `s₂` leaves and the pivot is `2`.

Iteration 2: a positive `1` remains under `x₃`. The `s₁` row has ratio `1/3`, and the `x₂` row has a negative coefficient. So `s₁` leaves and the pivot is `3`.

Optimum z = −16/3 = −5.33 at $x_1 = 0$, $x_2 = 8/3 = 2.67$, $x_3 = 1/3 = 0.33$

Check: z = 0 − 2(8/3) = −16/3

The same tableaus in decimals (as in the answer key)

Basic	x_1	x_2	x_3	s_1	s_2	Sol.
Z	−1	2	0	0	0	0
s_1	2	0	3	1	0	1
s_2	3	**2**	−1	0	1	5
Z	−4	0	1	0	−1	−5
s_1	2	0	**3**	1	0	1
x_2	1.5	1	−0.5	0	0.5	2.5
Z	−4.67	0	0	−0.33	−1	−5.33
x_3	0.67	0	1	0.33	0	0.33
x_2	1.83	1	0	0.17	0.5	2.67

First table 2m (every row 0.5m + 0.5m for pivot element).

Second and third table 3m (1m for every row). Every mistake in calculation −0.5m.